The Paleomagnetism of the Modipe Gabbro

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Samples have been collected from eleven sites in the Modipe gabbro which crops out on the border of Bechuanaland and South Africa. After partial demagnetization in alternating magnetic fields, ten of these sites give almost vertical directions of magnetization, forming a close group which corresponds to a paleomagnetic pole at 33°S, 31°E. The gabbro is intruded by the Gaberones granite, and samples from two sites in the granite show almost horizontal magnetization. On initial measurement gabbro samples collected near the contact yield directions close to those of the granite, which suggests that the gabbro was reheated at the time of emplacement of the granite (2350 m.y. ago). After magnetic cleaning, some of these samples give directions which agree with those at the other gabbro sites. Together with detailed alternating field and thermal demagnetization studies, this agreement provides strong evidence that the magnetically cleaned directions are those of the TRM acquired when the gabbro originally cooled. The gabbro has an age of approximately 2600 m.y. and is to date the oldest rock unit that has been studied paleomagnetically; the results imply the existence of a geomagnetic field in this early stage of the earth's history. Alternating field demagnetization experiments indicate that some of the magnetic grains have very high coercivities. Thermal demagnetization and IRM studies exclude the presence of hematite and show that the magnetic mineral present is magnetite. Microscopic observations and theoretical considerations suggest that the high coercivities observed are due to the shape anisotropy of small elongated single-domain grains of magnetite.

INTRODUCTION

The Modipe gabbro lies some 250 km northwest of Johannesburg at 24.7°S, 26.2°E. The intrusion is exposed over a distance of about 16 km, half of which lies in the Bechuanaland Protectorate (now called Botswana) 16 km east of Gaberones and half in the western Transvaal in the Republic of South Africa (Figure 1). It is to date the oldest rock unit that has been studied paleomagnetically.

The gabbro exhibits some interesting magnetic properties and despite its great age still contains a stable component of remanent magnetization which is believed to date from the time the rock originally cooled.

GEOLOGY

The gabbro is exposed as a line of isolated hills trending northwest-southwest. Apart from these hills exposure is very poor and the actual extent of the gabbro is not known; the reconnaissance map [*Jones*, 1963] is based to a large extent on variation of soil types (Figure 1). The gabbro is very uniform in composition, consisting almost entirely of plagioclase feldspar and augite with abundant accessory 'magnetite' [*Jones*, 1963].

To the northwest the gabbro is partly surrounded by the northeastern extremity of the Gaberones granite, the outcrop of which covers some 4500 km². On the northwestern slopes of Modipe Hill the gabbro is sheared and penetrated by veinlets of granitic material, and *Jones* [1963] therefore concludes that the Gaberones granite postdates the Modipe gabbro, although the actual contact is not exposed. The Bechuanaland Geological Survey regards the granite as post-Dominion Reef system and pre-Transvaal system in age. In terms of South African geochronology this places the granite between 2100 and 3000 m.y. old [*Nicolaysen*, 1962].

In adjacent parts of South Africa, about 20 km east of Wildebeest Kop, isolated outcrops of basic rocks along the Marico River have been correlated with the Modipe gabbro [*Poldervaart*, 1952]. *Schutte* [1963], in turn,

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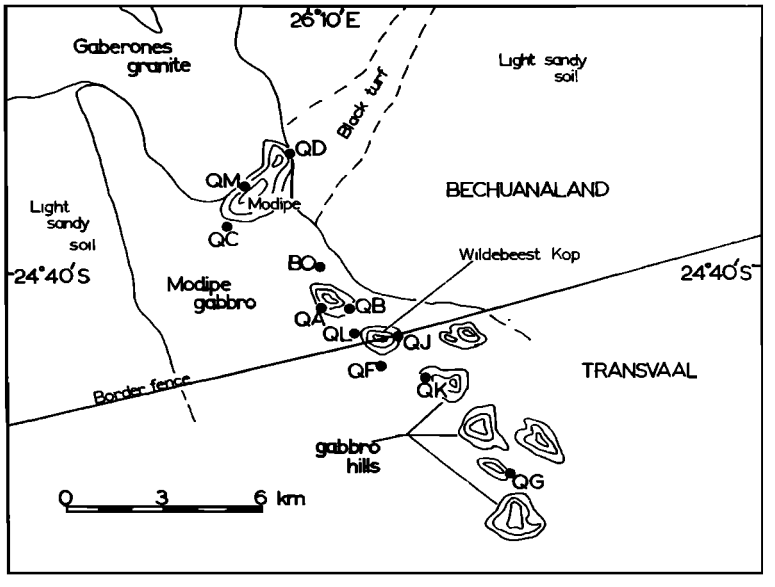


Fig. 1. Geological sketch map of the Modipe area showing the sampling sites.

correlates some of these Marico River intrusions with the Jamestown igneous complex, a very ancient Precambrian complex [*Nicolaysen*, 1962].

Radioisotopic age determinations for samples of the Gaberones granite and the Modipe gabbro indicate that the gabbro is probably greater than 2500 m.y. in age and may be as old as 3000 m.y. (Table 1).

Sampling and Measurement

Eleven sites were sampled in the Modipe gabbro (Figure 1), and two sites were sampled in the adjacent Gaberones granite batholith, 20 and 25 km, respectively, due west of Modipe Hill. At twelve of the thirteen sites, two cores, 60 cm long and 2.5 cm in diameter, were drilled with a modified version of the portable equipment described by *McElhinny and Neal* [1966]. The cores were oriented with respect to the sun's azimuth and the horizontal using the method of *Gough and Opdyke* [1963]. From the remaining site hand samples were collected which were oriented with a Brunton compass-clinometer.

From the oriented blocks, cores 2.5 cm in diameter were drilled in the laboratory. From each of these cores one or more cylinders 2.5 cm in length were cut. For the purposes of statistical analyses the mean of all the cylinders from a single block was regarded as a single observation (sample). From the 60-cm-long cores, two cylinders 2.5 cm in length were cut, one from the bottom and one from the 30-cm mark. It is believed that the distance of 30 cm separating these two cylinders justifies regard-

TABLE 1. Radioisotopic Ages of the Gaberones Granite and the Modipe Gabbro

Rock Unit	Method	Age, m.y.	Source
Gaberones granite	K-Ar on biotite	1629 ± 65	*Snelling* [1962]
Gaberones granite	K-Ar on hornblende	2185	Bechuanaland Geological Survey Annual Report for 1964 (N. J. Snelling, analyst)
Gaberones granite	K-Ar on hornblende	2225	
Gaberones granite	Rb-Sr whole rock	2820	
Gaberones granite	Rb-Sr whole rock	2655	
Gaberones granite	Rb-Sr whole rock	2340 ± 50	*McElhinny* [1966*b*]
Modipe gabbro	Rb-Sr whole rock	2630 ± 470	*McElhinny* [1966*b*]

ing them as two separate samples for the analyses.

The direction and intensity of natural remanent magnetization (NRM) were measured with a spinner magnetometer [*Gough*, 1964]. Specimens from all sites were subjected to alternating field demagnetization using the apparatus described by *McElhinny* [1966*a*] and following the criteria for magnetic cleaning outlined by *McElhinny and Gough* [1963].

The Modipe Gabbro

The site mean directions of magnetization before and after magnetic cleaning are given in Table 2. The NRM directions were generally widely scattered, and at three sites the resultant vector R is less than the value R_0 tabulated by *Watson* [1956] and *Irving* [1964], which is exceeded in only 5% of cases by a set of randomly oriented vectors.

Specimens from ten of the eleven sites all reached stable end points after alternating field demagnetization in the peak fields indicated in Table 2. At site QG, however, all the specimens were very weakly magnetized, and upon demagnetization their moments fell below the lower limit of the magnetometer before any stable end point had been reached. At the limit of measurement the magnetic vectors were approaching those of the group formed by the other ten sites. The specimens from this site show extensive alteration in thin section, whereas all the other sites appear to be reasonably fresh. Site QG has therefore been excluded from the following calculations. The ten site mean directions of magnetization form a good group almost vertically magnetized (Figure 2). Three of the sites (QC, QD, QM) have normal polarity (the same sense as the present earth's field) and seven are reversed. The mean normal and reversed directions differ by 170° ± 15°, and the mean direction of magnetization of the gabbro has therefore been calculated by reversing the unit vectors at the three normal sites and combining them with the seven reversed sites. This mean direction (see Table 2) corresponds to a paleomagnetic pole position, calculated on the assumption of a geocentric dipole field, at 33.3°S, 31.1°E, with oval of confidence defined by $\delta m = 10.8°$ and $\delta p = 10.7°$ [*Irving*, 1956]. Alternatively, the ten site poles may be combined to give the mean pole position, which then lies at 32.8°S, 30.9°E, with $R = 9.597$, $K = 22.3$, and $A_{95} = 10.5°$. This paleomagnetic pole is compared in Figure 3 with those of the Great Dyke, age 2530 ± 30 m.y. [*McElhinny and Gough*, 1963], and the Bushveld gabbro, age 1970 ± 70 m.y. [*Gough and van Niekerk*, 1959].

The angular dispersion given by $\Delta = 81/k^{1/2}$ for the site mean directions at Modipe is 9.1°, which is in close agreement with that calculated by *Creer* [1962] for the present earth's field at latitude 84° (the paleolatitude of Modipe).

It will be noticed in Figure 1 that the two polarity groups fall into two geographically distinct areas. The normal polarity sites (QC, QD, QM) are located at the northwest extremity of the outcrop around Modipe Hill, whereas the reversed sites all lie toward the southeast. This dual polarity can be explained by supposing that a reversal of the earth's magnetic field occurred while the Curie point isotherm passed from one group of sites to the other. Assuming that the gabbro was intruded at 1000°C into country rock at 0°C, the normal sites would have cooled to 600°C shortly after intrusion, whereas the reversed sites would have taken the order of 10^5 years to reach this temperature [*Jaeger*, 1961]. This is a minimum estimate based upon the present dimensions of the gabbro, and it is obviously very tentative because the original size and shape of the intrusion is not known. However, 10^5 years is probably more than sufficient time for the geomagnetic field to reverse itself [*Cox et al.*, 1964; *van Zijl et al.*, 1962; *Jacobs et al.*, 1959]. Self-reversal cannot be ruled out completely, but there is no evidence for it in any of the demagnetization data of NRM or TRM.

The Gaberones Granite

The Gaberones granite crops out as four large plutons [*Poldervaart*, 1952], the most northerly of which partly surrounds the Modipe gabbro. The NRM directions of the specimens from the two sites in the granite were scattered. Upon partial alternating field (af) demagnetization, several of the specimens reached similar stable end points, but in other cases, although the magnetic vectors were approaching that of the stable group, their moments fell

TABLE 2. Site Directions for the Modipe Gabbro

Site Name	Site Position Lat. S	Long. E		M_n D	M_n I	M_n R	M_n k	B_{ac}, mw m^{-2}	M_{ac} D	M_{ac} I	M_{ac} R	M_{ac} k	M_{ac} α_{95}	Site Pole Lat. °S	Site Pole Long. °E	Site Pole N or S
QA	24°40′	26°10′	2	294	25½	1.925	13	30	203	82½	1.999	1400	6.7	38.1	18.9	N
QB	24°41′	26°11′	4	332½	70	3.971	104	30–40	325½	84½	3 975	120	8.5	16.0	20.0	N
QC	24°39′	26° 9′	4	10	20½	3.901	30	30–40	286	−82½	3.944	54	12.7	27.8	42.1	S
QD	24°38′	26°10′	4	*		2.489		10–20	344½	−72	3.982	167	7.1	55.4	41.1	S
QF	24°41′	26°11′	4	180	83	3.957	70	20	3½	89	3.995	653	3.6	23.0	26 3	N
QJ	24°41′	26°12′	4	102	31	3.543	7	10–20	142	80½	3.989	283	5.5	38.5	40.4	N
QK	24°42′	26°12′	4	291	51½	3.809	16	20	318½	83	3.995	653	3.6	13.8	16.5	N
QL	24°41′	26°11′	3	*		2.257		20–30	211½	86	2.998	994	3.9	31.5	21.3	N
QM	24°39′	26° 9′	3	*		1.943		80–100	283	−80	2.955	45	18.7	27.6	47.6	S
BO	24°40′	26°10′	4	153	69	3.873	24	20	158	74	3.999	2900	1.7	51.7	43.9	N
Mean of ten site directions									154½	85	9.887	80	5.4	33.3	31.1	

N, number of samples measured.
R, resultant of the N unit vectors.
D, declination in degrees east of north.
I, inclination in degrees (positive downward).
M_n, total NRM.
M_{ac}, NRM after demagnetization in alternating field B_{ac}.
k, best estimate of precision [*Fisher*, 1953].
α_{95}, semi-angle of cone of 95% confidence for mean direction.
*, directions are random on *Watson's* [1956] criterion.

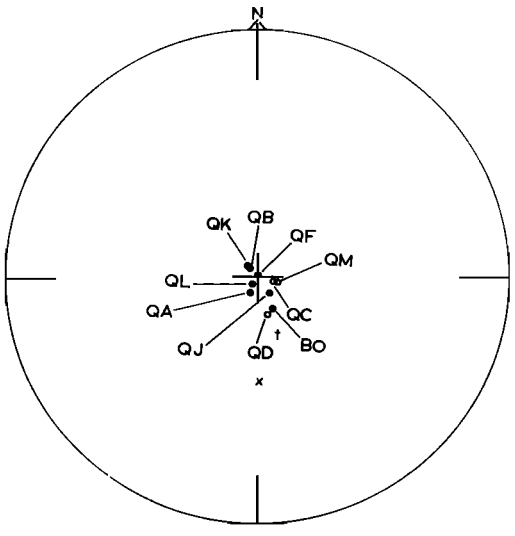


Fig. 2. Site mean directions of magnetization for ten sites in the Modipe gabbro. Lower-hemisphere stereographic projection. ● north-seeking direction down, ○ south-seeking direction down, × axial dipole field, south-seeking, † present field, south-seeking.

below the lower limit of the magnetometer before reaching it. The moments were variable, probably owing to varying amounts of magnetic minerals in each, and it was found necessary to cut more specimens from each 60-cm core to find those having large enough moments.

The directions of eleven specimens, which could be measured after magnetic cleaning in a peak field of 10 mw m^{-2}, are illustrated in Figure 4*a*, and their mean direction of magnetization is $D = 60°$, $I = 14°$, which is about 75° from that of the gabbro. This is only a preliminary result for the granite, and a further collection is to be made.

Reheating of the Gabbro by the Granite at Site QM

Site QM in the Modipe gabbro is estimated to lie 100 meters from the unexposed contact with the Gaberones granite. The NRM directions of three of the four hand samples (QM2, 3, 4) lie close to the granite directions (Figure 4*a*), but on partial af demagnetization the directions of two of these (QM2, 4) move steadily along great circles until they reach the stable direction of magnetization of the other gabbro sites. Of the remaining two samples, one (QM3) remains in good agreement with the granite directions throughout demagnetization and the other (QM1) remains close to the gabbro direction (Figure 4*b*).

The agreement of three of the NRM directions with that of the granite seems to indicate that the samples at this site must have been reheated at the time of emplacement of the granite and acquired a secondary component parallel to the field at that time. Specimens from three of the four samples were thermally demagnetized (see Figure 5) in a nonmagnetic furnace [*Opdyke,* 1964] by successively heating to higher temperatures and cooling in zero field. Temperatures could be measured to an accuracy of between 5 and 10°C. The direction of magnetization of QM2 remains unchanged after heating to 570°C but begins to move toward the gabbro stable direction after heating to 580°C. At this point the moment is barely measurable, and the primary component could not be detected by thermal demagnetization, although specimens from the same hand sample could be cleaned in a peak alternating magnetic field of 100 mw m^{-2}. The residual primary component after af demagnetization was only 0.5% of the NRM value, and it is not surprising that this could not be detected by thermal demagnetization because it is of comparable magnitude to the partial TRM

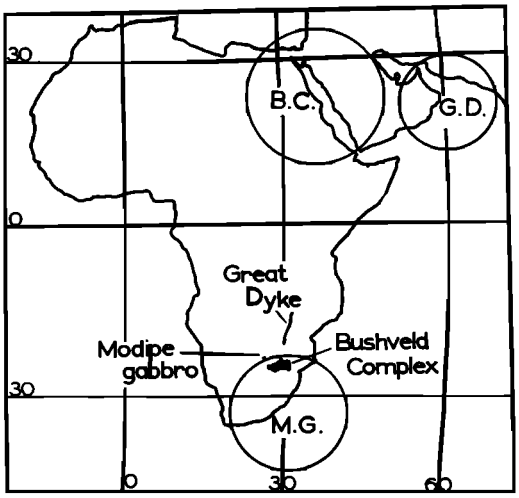


Fig. 3. Pole positions and circles of 95% confidence for the Modipe gabbro (M.G.), the Great Dyke (G.D.), and the Bushveld complex (B.C.).

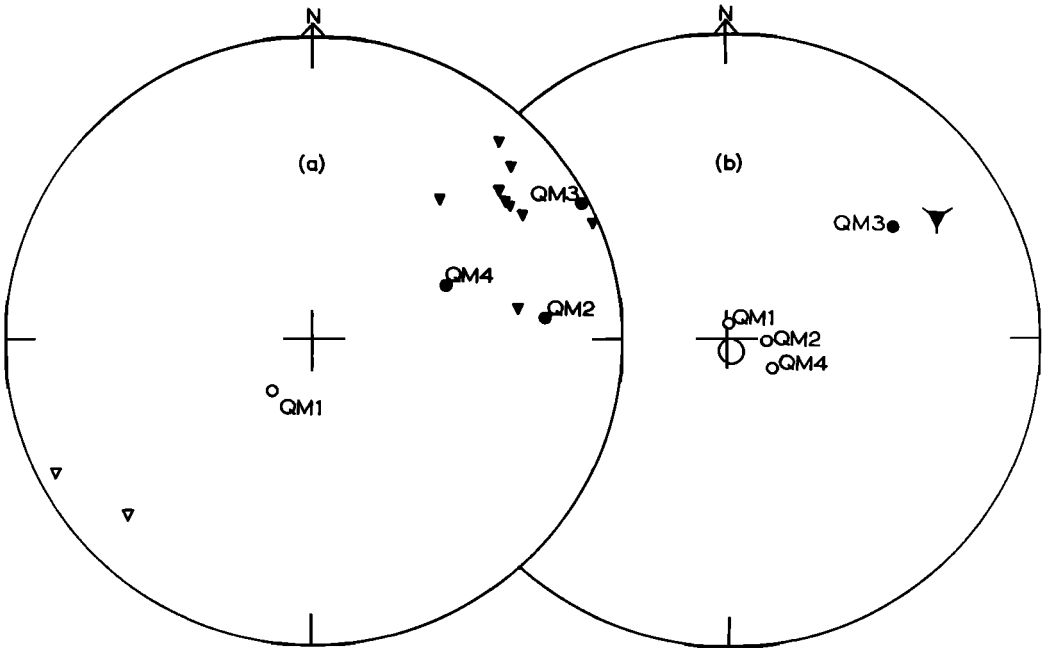


Fig. 4. (*a*) Directions of magnetization of eleven specimens of Gaberones granite after magnetic cleaning in 10 mw m^{-2} (triangles), and directions of NRM of the four Modipe gabbro samples from site QM (circles). (*b*) Directions of magnetization of the gabbro samples after magnetic cleaning in 80 mw m^{-2} (QM1, 4) and 100 mw m^{-2} (QM2, 3). Also shown are the circle of 95% confidence of the gabbro mean direction (large circle) and the Gaberones granite mean direction (large triangle). Lower-hemisphere stereographic projection; open symbols, south-seeking direction; solid symbols, north-seeking direction.

produced by stray fields in the furnace ($\approx$10 γ). The directions of magnetization of QM1 and 3, however, remain close to their NRM directions, as was observed during af demagnetization.

Age of the Primary Magnetization

The measurements of samples from site QM suggest that the gabbro there was reheated to within a few degrees of its Curie point, and they indicate that this rock can retain a partial TRM for at least 2×10^9 years. There seems little doubt, therefore, that the stable magnetization observed at the other gabbro sites is the TRM acquired when the rock originally cooled. This is perhaps the most significant result of the investigation; it implies that a geomagnetic field existed at the time of intrusion of the Modipe gabbro, approximately 2600 m.y. ago.

The earth's magnetic field originates in the core, and, if the core has not always been its present size, at this stage in the history of the earth it must have been large enough to permit the fluid motions necessary to maintain a geomagnetic field.

Magnetic Properties of the Gabbro

Alternating field demagnetization. Most of the specimens from the Modipe gabbro had acquired a secondary component of magnetization which could be removed by magnetic cleaning. Directions of magnetization at site BO, however, remained unchanged after magnetic cleaning, which suggested that little or no secondary component was present. Specimens from this site were therefore the most appropriate to use in an investigation of the magnetic properties of the stable component in the gabbro.

Upon af demagnetization the magnetic moments of these specimens fall off exponentially with increasing field (Figure 6); after 100 mw m^{-2}, 10% remains. At 180 mw m^{-2} only

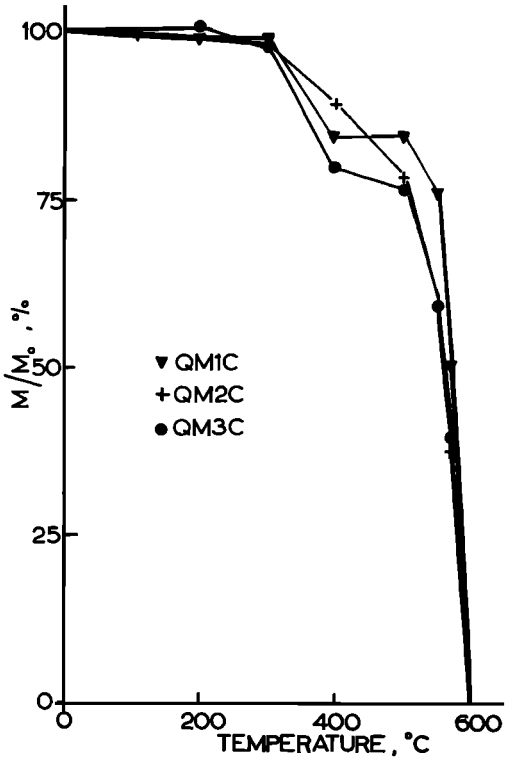


Fig. 5. Thermal demagnetization curves for three samples from site QM.

1% is left, and the direction is still unchanged. This gradual decrease in magnetic moment demonstrates that the gabbro has a very broad coercivity spectrum, which presumably indicates a large range in grain and domain size [*Stacey*, 1963]. Similar high coercivities in coarse-grained dolerites have been reported by *McElhinny and Opdyke* [1964].

The same specimens were then heated to 700°C and cooled in the earth's magnetic field, and the TRM acquired in each case was demagnetized. The demagnetization curves for the NRM and TRM of one of these specimens are shown in Figure 6. The great similarity of these curves demonstrates that the NRM and TRM have almost identical coercivity spectrums. This provides further evidence that the NRM is a primary TRM and suggests that heating these specimens to 700°C in air caused little or no alteration.

High coercivities (>100 mw m^{-2}) are usually attributed to hematite [e.g., *Creer*, 1959], but this mineral is rare as an original constituent of igneous rocks [*Kerr*, 1959]. We therefore attempted to determine from thermal demagnetization and isothermal remanent magnetization experiments how much hematite, if any, is present.

Thermal demagnetization. Figure 7*a* shows that the thermal demagnetization curves of the NRM and TRM of specimen BO2A are almost identical, which again indicates that the NRM is a primary TRM and that heating to 700°C caused very little, if any, alteration. The curves show a very sharp decrease between 550 and 600°C, in which range 94% of the initial moment is destroyed. After heating to 600°C, 0.4% of the NRM and 1.0% of the TRM are left. Similar results were obtained for specimen BO1A, and 0.3% of the NRM and 0.5% of the TRM remained at this temperature. No systematic direction could be observed after heating to 600°C. Thus not more than 1% of the remanence can be attributed to a mineral having blocking temperatures in excess of 600°C, whereas grains with coercivities greater than 100 mw m^{-2} account for 10%.

As a further check, two specimens were first demagnetized in fields of 100 and 150 mw m^{-2} peak value, respectively, and the high-coercivity remanences which were not removed by this treatment (12 and 4% of the NRM, re-

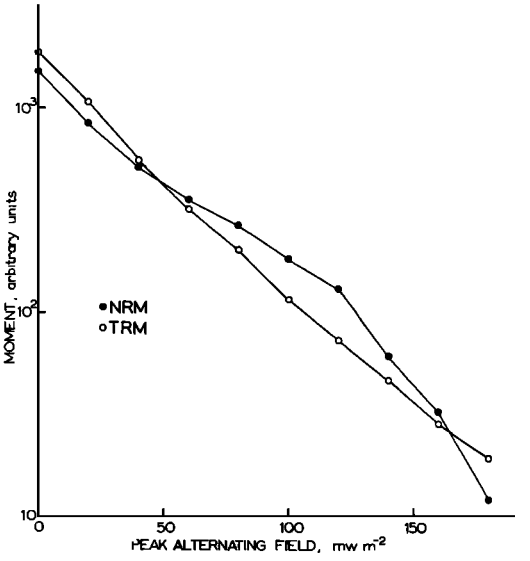


Fig. 6. Alternating field demagnetization curves of NRM and TRM of specimen BO1A2.

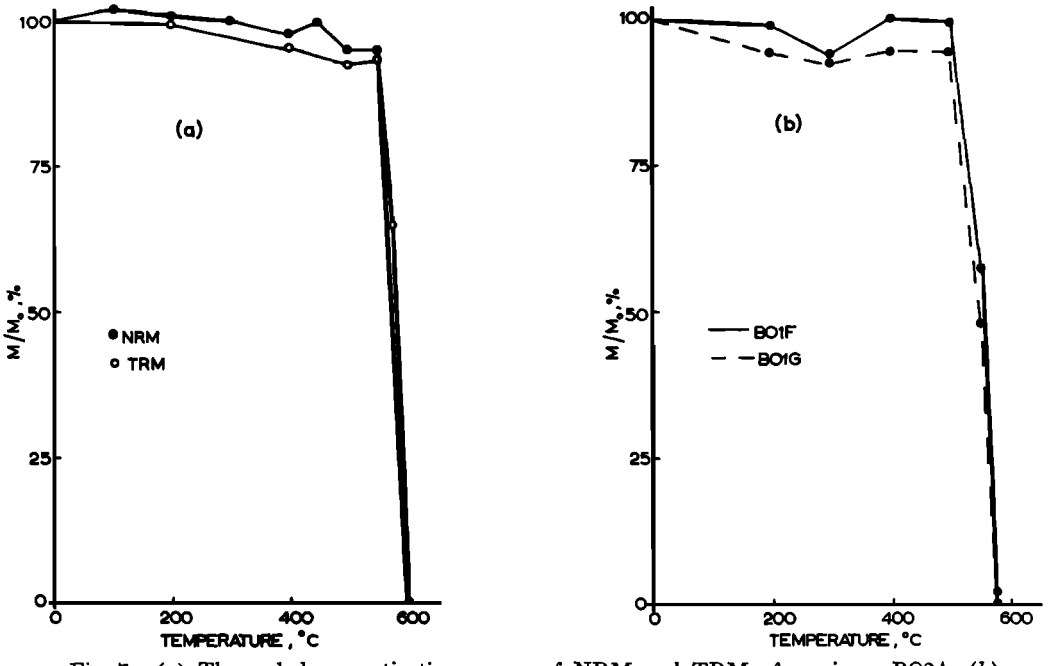


Fig. 7. (*a*) Thermal demagnetization curves of NRM and TRM of specimen BO2A. (*b*) Thermal demagnetization curves of the NRM's remaining in specimens BO1F and BO1G after alternating field demagnetization in 100 and 150 mw m^{-2}, respectively.

spectively) were then thermally demagnetized (Figure 7*b*). Once again, the sharp decrease in moment is observed between 550 and 600°C, at which point the moments of both specimens were barely measurable.

All the remanence in these specimens resides in grains having critical blocking temperatures less than 600°C, and thus the thermal demagnetization studies indicate that the magnetic mineral present is either magnetite (Curie point 580°C) or a member of the ilmenite-hematite solid solution series (approximately 10% ilmenite).

Isothermal remanent magnetization. Roquet [1954] has studied the isothermal remanent magnetization (IRM) of magnetite and hematite samples of various grain sizes. Her results show that magnetite is saturated after being placed in a field of 0.3 w m^{-2} (3 kilogauss) and hematite approaches saturation only in fields of 3 w m^{-2}. The hematite curves start to flatten out above 2 w m^{-2}, and the main rise in the IRM occurs between about 0.4 and 2 w m^{-2}. If hematite were present in appreciable amounts in a specimen whose magnetic mineral is predominantly magnetite, one would expect to see a rise in the IRM of the specimen between 0.4 and 2 w m^{-2} after the magnetite has been saturated.

Specimen BO2E1 (volume 7.22 cm^3, weight 17.4 g) was first demagnetized in a peak alternating magnetic field of 180 mw m^{-2} and its IRM measured after being placed successively in fields of up to 1.8 w m^{-2} (the maximum available). The magnetometer was carefully calibrated before and after each step to improve the accuracy of measurement of the moment acquired in each case. In this way the moments were obtained to an accuracy of 1%, instead of 5% as obtained from routine measurements. A plot of the IRM acquired in various fields is shown in Figure 8. The specimen is saturated in a field of 0.3 w m^{-2}, and its moment of 8.57×10^{-5} amp m^2 (8.57×10^{-2} emu) remains unchanged up to 1.8 w m^{-2}. This saturation moment is the mean of all the five IRM's acquired in fields above 0.3 w m^{-2} and has standard deviation 0.06×10^{-5} amp m^2—that is, about 1% of the mean value—which corresponds to the accuracy claimed for moments on the magnetometer. Thus any rise in the IRM between 0.4 and 1.8 w m^{-2} cannot

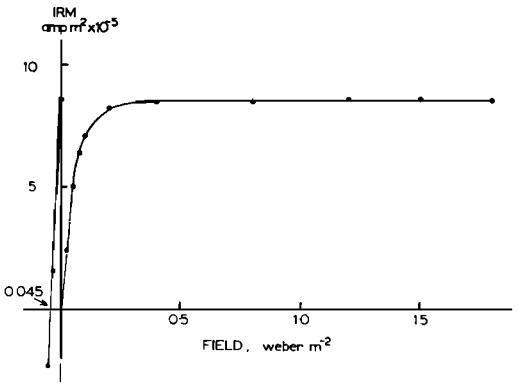


Fig. 8. IRM of specimen BO2E1.

exceed 1% of the value at 0.4 w m^{-2}, and therefore any hematite present cannot contribute more than 1% to the IRM in 1.8 w m^{-2}.

In addition to IRM studies of hematite, *Roquet* [1954] has investigated the TRM acquired by the same samples in various fields. Her results for two different grain sizes are summarized in Table 3. *Irving* [1957] has separated specularite of much larger grain size and reports IRM values in fields of up to 0.8 w m^{-2}. His values up to this limit are close to those of Roquet's sample F_5, which has the larger of the two grain sizes investigated. In our sample BO2E1 the IRM moment due to hematite acquired in 1.8 w m^{-2} cannot exceed 8×10^{-7} amp m^2 (1% of the total IRM); this corresponds to 0.01 gram of hematite using Roquet's F_4 data and 0.003 gram using her F_5 data. In a field of 3×10^4 γ our sample acquired a TRM moment of 1.5×10^{-5} amp m^2. Using Roquet's TRM data we find that 0.008% (F_4 data) or 0.36% (F_5 data) of the TRM in our sample can be due to hematite. Both these values are less than the 1% of the TRM of our samples which remains after af demagnetization in a peak field of 180 mw m^{-2}. We conclude that hematite cannot account for any of the high coercivity remanence up to 180 mw m^{-2} in the Modipe gabbro. The high coercivities must therefore be attributed to magnetite.

The 'back' field (B_{cr}) required to destroy the saturation IRM (Figure 8) is 45 mw m^{-2} (450 gauss), which is a typical value for stable igneous rocks [*Irving*, 1964].

Microscopic investigations. Examination of thin sections shows that the opaque minerals occur in two distinct forms which almost certainly represent a primary and a secondary phase. The primary opaques occur as elongated rods of varying length exsolved on the cleavage planes of the clinopyroxene; the secondary phase consists of irregularly shaped but generally equidimensional grains associated with chlorite.

Under reflected light the larger of the primary opaques are seen to be magnetite, but many of the exsolved 'rods' are too small to be clearly identified. The secondary opaques are mainly leucoxene after ilmenite, sometimes containing small grains of magnetite.

Origin of the High-Coercivity Remanence

Verhoogen [1959] has suggested that the remanence of small regions where mechanical stresses are associated with lattice dislocations may be extremely resistant to af demagnetization. His theory predicts that stresses of 10^9 newtons m^{-2} (10^{10} dynes cm^{-2}) are necessary to account for coercivities of 150 mw m^{-2} or more, as we have observed. Such large internal stresses are close to the upper limit suggested by *Stoner and Wohlfarth* [1947], and the probability that they would occur frequently enough to explain 5 to 10% of the remanence of a large plutonic intrusion, which cooled very slowly, must be extremely small.

Stoner and Wohlfarth point out that with coercivities greater than 50 mw m^{-2} there is strong presumptive evidence of some degree of shape anisotropy. Microscopic examination of our samples shows that the primary opaque mineral occurs as elongated grains, some of which have axial ratios as high as 1:20 and lengths of a few microns. *Morrish and Yu* [1955] have shown that magnetite grains of axial ratio 1:10 will be single-domained if they are 2 to 3 μ long, but they regard this as a minimum estimate. It therefore seems rea-

TABLE 3. IRM and TRM of Hematite
[after *Roquet*, 1954]
Values are given in amp m^2 per grain.

Sample	Grain Size, microns	IRM in 1.8 w m^{-2}	TRM in 3×10^4 γ
F_4	0.1	8.0×10^{-5}	1.2×10^{-7}
F_5	0.25	2.4×10^{-4}	1.8×10^{-5}

sonable to assume that some of the elongated magnetite grains in the Modipe gabbro are single-domained. If this is so, the microscopic coercive force of grains of a given shape can be calculated from theory, but the microscopic coercive force, B_c, of small single-domain grains is always greater than the peak alternating magnetic field, B_B, required to 'unblock' them [*Néel*, 1949; *Dunlop*, 1965]. However, for the grain volumes in question (approximately 10^{-11} cm^3) $B_c \approx B_B$, and the calculated microscopic coercive force can be taken as a close estimate of the peak alternating magnetic field which these grains can withstand.

For a single-domain grain we have

$$B_c = \mu_0 M_s \Delta N$$

where μ_0 is the permeability of free space ($4\pi \times 10^{-7}$ henry m^{-1}), M_s is the saturation magnetization, and ΔN is the difference in demagnetizing factors perpendicular to and parallel to the long axis. For magnetite $M_s \approx 4 \times 10^5$ amp m^{-1} (4×10^2 emu cm^{-3}) and grains with axial ratio 1:10 or more have $\Delta N = 0.5$ (2π in unrationalized units). From these values we obtain $B_c \approx 250$ mw m^{-2}, which is greater than the maximum peak alternating magnetic field used in this work. It appears, therefore, that coercivities as high as 250 mw m^{-2} can be explained in terms of the shape anisotropy of small elongated single-domain grains of magnetite.

In several studies made in this laboratory it has been observed that some gabbros and coarse dolerites are extremely resistant to af demagnetization. In contrast to lava flows and fine-grained intrusives, such hypabyssal and plutonic rocks cool very slowly and are unlikely to contain large internal stresses. Thus, although the theory of *Verhoogen* [1959] may account for some high coercivities observed in fine-grained rocks, it is unlikely to apply to coarse-grained rocks in which high coercivities are more readily explained in terms of the shape anisotropy of small grains.

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